

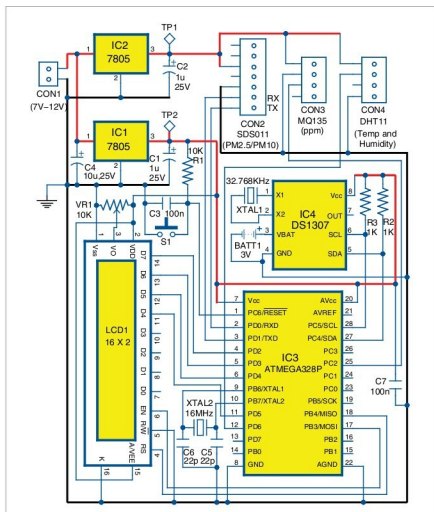


Fig. 2: Circuit diagram of the air-quality meter

the circuit is ATmega328P (IC3). Other components used are voltage regulators 7805 (IC1 and IC2), 16 × 2 LCD display (LCD1), temperature and humidity module DHT11 (connected to CON4), gas sensor MQ135 (connected to CON3), PM2.5/PM10 sensor (connected to CON2) and a few others.

ATmega328P. It can be programmed with embedded software using a standard programmer or Arduino IDE. ATmega328P offers 23 input/output functional ports, and a 16MHz crystal oscillator is used to provide timing/clock reference.

Temperature and humidity module (DHT11). This composite sensor contains a calibrated digital signal output of temperature and

humidity. The sensor (connected to CON4) includes a resistive-type humidity measurement component and an NTC temperature-measurement device. Its output pin is connected to pin 12 of IC3.

Serial real-time clock (RTC) IC (DS1307). It is a low-power, full binary-coded decimal (BCD) clock/calendar with 56 bytes of NV SRAM. Addresses and data are transferred serially through an I2C, bi-directional bus. The clock/calendar provides information about seconds, minutes, hours, days, dates, months and years. SCL and SDA pins of IC4 are connected to SCL (pin 28) and SDA (pin 27) pins of IC3, respectively.

Air-quality sensor (MQ135). The sensitive material of the sensor is

PARTS LIST

Semiconductors:

- IC1, IC2 - 7805, 5V regulator
- IC3 - ATmega328P MCU with Arduino Uno bootloader
- IC4 - DS1307 serial, I2C real-time clock

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 10-kilo-ohm
- R2, R3 - 1-kilo-ohm
- VR1 - 10-kilo-ohm potentiometer

Capacitors:

- C1, C2 - 1µF, 25V electrolytic
- C3, C7 - 100nF ceramic disk
- C4 - 10µF, 25V electrolytic
- C5, C6 - 22pF ceramic disk

Miscellaneous:

- XTAL1 - 32.768kHz crystal
- XTAL2 - 16MHz crystal
- BATT1 - 3V battery
- S1 - Tactile switch
- LCD1 - 16×2 character module display
- SDS011 - PM2.5/PM10 sensor
- MQ135 - ppm gas sensor
- DHT11 - Temperature/humidity sensor
- CON1 - 2-pin connector terminal
- CON2 - 7-pin connector
- CON3, CON4 - 4-pin connector
- 12V DC supply/12V battery
- 7805 heat-sink

tin-dioxide (SnO₂), whose conductivity increases with concentration of gas. Change in conductivity is converted into output voltage signal, which varies corresponding to the concentration of the combustible gas. MQ135 is highly sensitive to ammonia, sulphide and benzene steam, smoke and other harmful gases. It is a low-cost sensor, suitable for different applications. Output of the gas sensor (CON3) is connected to analogue input pin 25 of IC3.

PM2.5/PM10 sensor (SDS011). SDS011 is an air-quality sensor developed by Inovafit, a spin-off from University of Jinan (Shandong). Because of its size, it is probably one of the best sensors in terms of accuracy. While other sensors tend to focus on shrinking the sensor size, SDS011 has opted for a size tradeoff, allowing it to be used as a larger fan—larger the fan, better the quality.

SDS011 uses the principle of laser scattering in the air, which can be obtained from 0.3 microns to 10 microns suspended particulate matter concentration. Data is stable and reliable, and there is a built-in fan, digital output and a high degree of integra-